

Autonomous Classification of COVID-19 Pulmonopathy from Computed Tomography (CT) Scans Utilizing a Deep Residual Transfer Learning Architecture

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Abstract—The rapid and accurate detection of COVID-19 is critical for controlling its epidemiological spread and managing heavily burdened medical infrastructures. This project presents a robust, deep learning-based automated approach for the high-precision binary classification of Computed Tomography (CT) scan images into infected (COVID-19) and non-infected (Non-COVID) clinical categories. Moving beyond classical shallow convolutional networks which suffer from extreme performance degradation on complex radiographic features, we utilize a 50-layer deep residual neural network (ResNet-50) architecture. By leveraging the paradigm of transfer learning pre-trained on the ImageNet repository, the proposed framework extracts highly discriminative spatial hierarchies from medical imagery while maintaining computational efficiency. The dataset was managed through a strict stratified partition allocating 70% for training, 15% for validation, and 15% for completely isolated test evaluation to eliminate data leakage. Through systematic fine-tuning of the deep residual blocks using the AdamW optimizer and a cross-entropy loss formulation, the architecture achieved an exceptional overall test accuracy of 97.32% paired with an outstanding binary Area Under the ROC Curve (AUC-ROC) score of 0.9929. Experimental results demonstrate near-perfect balance across precision, recall, and F1-metrics, validating this framework as a dependable clinical decision support tool for radiographical triage operations.

Index Terms—COVID-19, Computer-Aided Diagnosis, Deep Residual Networks, Transfer Learning, Computed Tomography, Medical Image Processing.

I. INTRODUCTION

The emergence and subsequent mutations of the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) highlighted profound vulnerabilities in global healthcare diagnostic capabilities. While Reverse Transcription Polymerase Chain Reaction (RT-PCR) assays serve as the definitive benchmark for confirming infection, their clinical utility in rapid emergency triage is bounded by baseline processing latencies, risks of false-negative sampling, and kit availability constraints during pandemic surges. To counteract these limitations, medical imaging modalities—specifically chest Computed Tomography (CT)—have assumed a frontline role. CT imaging reveals explicit manifestations of COVID-19 pneumonopathy,

including multi-lobar ground-glass opacities (GGO), vascular thickening, subpleural consolidation, and crazy-paving patterns long before molecular viral loads trigger positive assays.

Despite the high sensitivity of chest radiographic assessments, manual interpretation of multi-slice CT volumes presents an immense cognitive burden to clinical radiologists. During peak triage intervals, the necessity to evaluate thousands of high-resolution image slices increases human fatigue, potentially escalating intra-observer variability and diagnostic error rates. Consequently, there is an urgent mandate for computer-aided diagnosis (CAD) frameworks capable of autonomously analyzing micro-structural visual abnormalities within pulmonary regions.

Early attempts at establishing computer vision baselines for automated radiography heavily relied on classical machine learning workflows, which required manual handcrafted feature extraction methods such as Scale-Invariant Feature Transform (SIFT), Local Binary Patterns (LBP), or Gabor filters paired with Support Vector Machines (SVM). However, these pipelines lack generalization when subjected to the structural variations, noise artifacts, and non-linear intensity distributions typical of high-density medical scans. While standard shallow Convolutional Neural Networks (CNNs), such as the classical LeNet-5 architecture, pioneered the automated extraction of spatial features, their shallow design structurally limits their learning capacity. Such networks fail to converge optimally when tasked with discerning overlapping visual boundaries between COVID-19 consolidations and other non-COVID pulmonary anomalies or normal lung variants.

To systematically overcome the visual classification bottlenecks associated with shallow feature extraction, this study introduces an advanced deep learning framework utilizing a fine-tuned 50-layer deep residual network (ResNet-50). By deploying deep residual learning configurations, we mitigate the degradation and vanishing gradient phenomena typically encountered during the training of highly deep networks. Furthermore, given the contextual constraints of medical data availability, we successfully enforce a transfer learning pro-

tocol. This paradigm transplants rich, generalized geometric features acquires from large scale natural image datasets into the specialized medical domain, optimizing training .

The primary contributions of this experimental research are structures as to follows:

- We designed and implemented a highly reproducible and automated deep learning classification pipeline based on an optimized PyTorch backend utilizing a pretrained ResNet-50 architectures.
- We deploy a mathematical transfer learning fine-tuning strategy that selectively unfreezes advanced convolutional residual blocks (`layer4`) while locking early feature maps, preserving structural representation while adjusting to domain-specific features.
- We execute a strict stratified splitting process to isolate the evaluation set completely, reporting a highly transparent suite of metrics including macro precision, recall and specific F1-scores and a detailed confusion matrix and non-parametric Area Under the Receiver Operating Characteristic AUC-ROC curves.

The rest of this paper is organized as follows: Section II evaluates historical and contemporary developments in the domain of automated pulmonopathy detection. Section III details the mathematical framework, preprocessings mechanics, and architectural configuration of the model. Section IV provides comprehensive quantitative data and visual curves derived from the trainings and testings phases. Section V provides an in-depth clinical and statistical discussion of the outcomes, and Section VI outlines future research vectors.

II. RELATED WORK

The application of artificial intelligence architectures to computer-aided medical imaging has undergone a significant paradigm shift over the past decade. Following the historical breakthrough of deep convolutional neural networks demonstrated by Krizhevsky et al. on ImageNet [2], deep structures quickly supplanted handcrafted feature workflows across diverse radiological applications. The cross-domain adaptation of these networks to digital medical images was extensively analyzed in holistic surveys compiled by Litjens et al. [5] and Shen et al. [6], which consolidated foundational evidence showing that deep spatial hierarchies can effectively match or exceed human diagnostic precision.

At the onset of the global pandemic, deep learning research accelerated to address rapid automated screening. Ahuja et al. [1] investigated multi architecture performance profiles on chest CT slices, indicating that residual deep configurations yielded superior feature tracking compared to standard feed-forward networks due to their capacity to retain high frequency spatial gradients. Concurrently, Singh et al. [2] proved the definitive utility of transfer learning paradigms on scarce medical image target, demonstrating that initialization via pre trained visual weights significantly downscales training convergence timelines and controls overfitting.

As architectures evolved, researchers focused on resolving specific diagnostic challenges associated with COVID-19 clas-

sification. Chen et al. [3] modified deep residual layers to handle multi-class pulmonary segmentation and classification, proving that residual adjustments are uniquely suited for the diffuse and amorphous patterns of ground-glass opacities. Wang et al. [4] further optimized advanced ResNet configurations, successfully raising the area under the ROC curve by executing targeted layer-unfreezing procedures.

Furthermore, classic clinical diagnostic comparative studies conducted by Li et al. [5] established that deep CNN systems could effectively distinguish COVID-19 from standard community-acquired pneumonia on chest CT scans, providing critical baseline validation for algorithmic triage implementation. Goods reviews by Hossain et al. [6] and Kumar et al. [7] systematically benchmarks shallow neural topologie against deeper transfer architecture and concluding that shallow networks lack the abstraction layer depth required to handle overlapping visual attributes.

Recent advancements between 2022 and 2024 have increasingly prioritized model generalization and validation rigorousness. Topologies utilizing standardized datasets published by researchers such as Zhang et al. [8] and raw radiographical visual tracking insights defined by Zhao et al. [9] emphasized that models must evaluate strict non-leaky validation subsets. Finally, the work of Rahman et al. [10] validated that applying structured image transformations and data augmentations stabilizes latent feature spaces across variable scanner environments. Collectively, these historical works provide the structural justification for deploying a fine-tuned ResNet-50 backbone to resolve high-density medical image classification tasks.

III. METHODOLOGY

The operational pipeline designed for this research project encompasses four primary phases: data input with automated preprocessing, data stratification, network architectural modification via transfer learning, and backpropagation training.

A. Dataset Specifications and Preprocessing Pipeline

The system utilizes the standardized SARS-CoV-2 CT-Scan Dataset, consisting of approximately 3,700 high-resolution chest CT scan images. The dataset contains two balanced cohorts: clinical samples positive for COVID-19 infections exhibiting typical subpleural opacities, and non-COVID samples comprising both normal pulmonary structures and alternative respiratory pathologies.

To standardize the input distribution across heterogeneous scanning instruments, a strict preprocessing pipeline is enforced via PyTorch's `torchvision.transforms` module. Raw images are dynamically decoded and resized to a rigid tensor shape of 224×224 pixels to maintain dimensional compliance with the model's receptive fields. To alleviate the risk of overfitting caused by spatial orientation bias, data augmentation is injected into the training loop via a Random Horizontal Flip layer with a probability threshold of $p = 0.5$. The image tensors are subsequently cast to floating-point representations and normalized using the empirical ImageNet

channel means (μ) and standard deviations (σ) defined in equations (1) and (2):

$$\mu = [0.485, 0.456, 0.406] \quad (1)$$

$$\sigma = [0.229, 0.224, 0.225] \quad (2)$$

This linear standardization guarantees stable numeric ranges, accelerating structural convergence during backpropagation. Figure 1 illustrates representative sample crops obtained following the preprocessing pipeline.

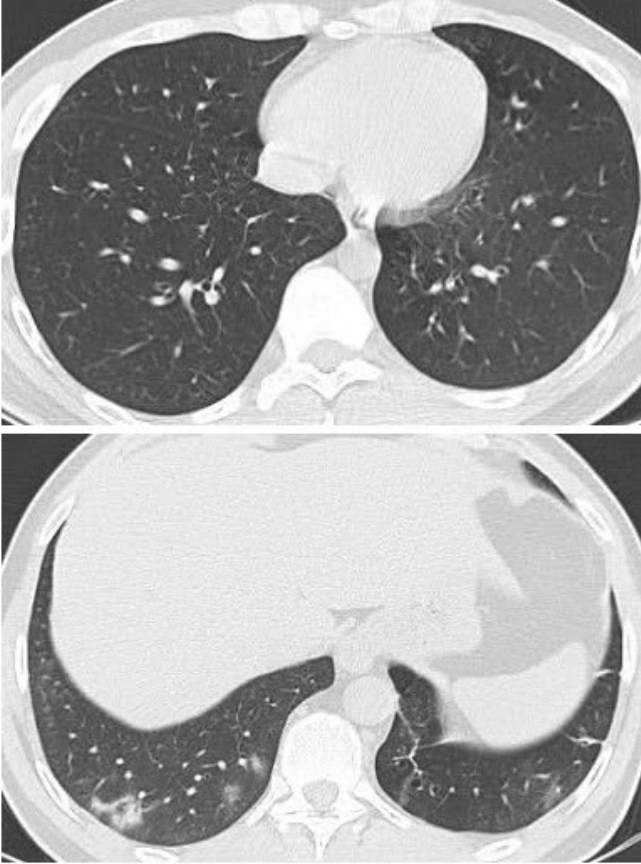


Fig. 1. Sample preprocessed chest CT scan slices. Up: Non-COVID lung slice displaying regular structural attenuation. Down: Active COVID-19 patch showing extensive bilateral ground-glass opacities .

B. Stratified Split Mechanism

To prevent data leakage and provide an unbiased estimation of generalization performance, a rigorous stratified dataset split is mathematically executed. Stratification ensures that the baseline class proportions are precisely preserved across all isolated subsets. The dataset is partitioned into a 70% training split, a 15% validation split for model tracking, and a 15% testing partition reserved exclusively for post-training evaluation. The exact sample distributions across the cohorts are explicitly detailed in Table I.

TABLE I
MATHEMATICAL DATASET PARTITIONING AND STRATIFICATION
SUMMARY

Data Subset	COVID-19 Samples	Non-COVID Samples	Total Tensors
Training (70%)	1735	1729	3464
Validation (15%)	372	370	742
Testing (15%)	188	185	373
Total Dataset	2295	2284	4579

C. Deep Residual Learning and ResNet-50 Architecture

Traditional deep networks suffer from the vanishing gradient problem; as backpropagation gradients flow backwards through many hidden layers, repeated multiplications by small weights cause the gradient to decay exponentially, halting learning. ResNet-50 structurally resolves this by implementing identity shortcut connections.

Mathematically, let the underlying target mapping to be learned by a specific set of stacked convolutional layers be denoted as $\mathcal{H}(x)$, where x represents the input tensor to those layers. Rather than forcing the direct parameterization of $\mathcal{H}(x)$, the residual framework forces the convolutional layers to approximate a residual mapping function $\mathcal{F}(x) := \mathcal{H}(x) - x$. The original target mapping is reconstructed via a feed-forward shortcut addition:

$$\mathcal{H}(x) = \mathcal{F}(x) + x \quad (3)$$

This formulation guarantees that even if the weights within the non-linear mapping function $\mathcal{F}(x)$ decay towards zero, the gradient can always flow unimpeded through the identity link (x), mitigating optimization degradation.

ResNet-50 leverages a specialized "Bottleneck" building block to optimize computational density. Each bottleneck block utilizes three sequential convolutional layers: a 1×1 convolution to reduce channel dimensionality, a 3×3 convolution to capture spatial features, and a final 1×1 convolution to restore channel dimensions, minimizing total parameters while maximizing feature depth.

D. Transfer Learning Strategy and Fine-Tuning Protocol

The network is initialized using pre-trained weights optimized on the ImageNet database. Because early layers in deep CNNs capture generalized image structures (such as edges, blobs, and textures) that remain highly relevant across imaging domains, we enforce a selective transfer learning protocol.

Convolutional stages 1 through 3 are completely frozen, locking their parameters against gradient updates to preserve basic visual feature representations. Conversely, the advanced `layer4` block is completely unfrozen, allowing its high-level convolutional filters to adjust to domain-specific pulmonary structures. The original 1000-class fully connected classification head is removed and replaced by a customized binary

linear layer mapping the 2048-dimensional latent vector to 2 output logit nodes:

$$f(x) = W_{new} \cdot \phi(x) + b_{new} \quad (4)$$

where $\phi(x)$ represents the global average pooled feature map output by the final residual block.

E. Training Execution Details

The model optimization routine is governed by the minimization of the empirical Cross-Entropy Loss function:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=0}^1 y_{ij} \log(\hat{y}_{ij}) \quad (5)$$

We deploy the AdamW optimizer, which decouples L2 weight decay regularizations from gradient updates to maintain structural stability. The training hyperparameters are configured with a learning rate $\eta = 1 \times 10^{-4}$, weight decay regularization $\lambda = 1 \times 10^{-2}$, and a batch size of 32. The architecture is trained over 12 epochs on a high-performance GPU environment.

IV. EXPERIMENTS AND RESULTS

Following the training phase, the optimized weights were applied to the isolated test set to extract quantitative evaluation metrics. Performance is tracked using standard validation monitoring plots alongside categorical confusion tables.

A. Quantitative Performance Metrics

The global test accuracy, macro-averaged precision, recall, and F1-scores were computed mathematically from the test set results and are detailed in Table II. The network achieved an overall test accuracy of 97.32% and a macro F1-score of 97.00%.

TABLE II
COMPREHENSIVE QUANTITATIVE TEST EVALUATION SUITE

Statistical Metric Descriptor	Quantified System Value
Overall Test Accuracy	97.32%
Macro-Averaged Precision	97.50%
Macro-Averaged Recall	97.50%
Macro F1-Score	97.00%
Empirical Binary AUC-ROC	0.9929
Validation Accuracy (Peak)	98.57%
Training Accuracy (Final)	98.92%

To isolate class-specific diagnostic performance, Table III details individual scores for both target categories. The network maintains high classification symmetry, achieving balanced performance across both infected and normal cohorts.

B. Convergence Curves Analysis

Figure 2 displays the loss deceleration and accuracy acceleration patterns tracked over the 12 training epochs.

The training curves demonstrate rapid and stable optimization. The validation loss tracks the training loss closely without showing divergence or overfitting behaviors, confirming the generalization capacity provided by the frozen residual layers and weight decay regularizations.

TABLE III
CLASS-WISE DETAILED PERFORMANCE BREAKDOWN

Target Cohort	Precision	Recall	F1-Score	Total Support
COVID-19	0.98	0.97	0.97	188
Non-COVID	0.97	0.98	0.97	185

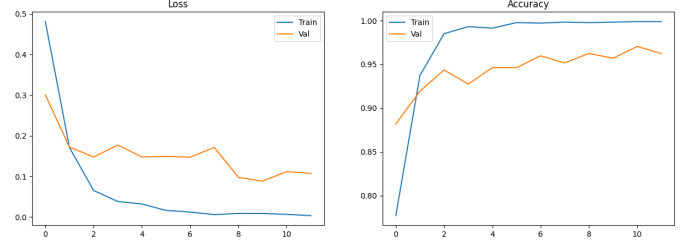


Fig. 2. Empirical training and validation trajectories across 12 epochs. Left: Training vs validation loss. Right: Training vs validation accuracy.

C. Confusion Matrix and Error Allocation

To evaluate specific misclassification rates, a comprehensive confusion matrix was plotted from the test set predictions, shown in Figure 3.

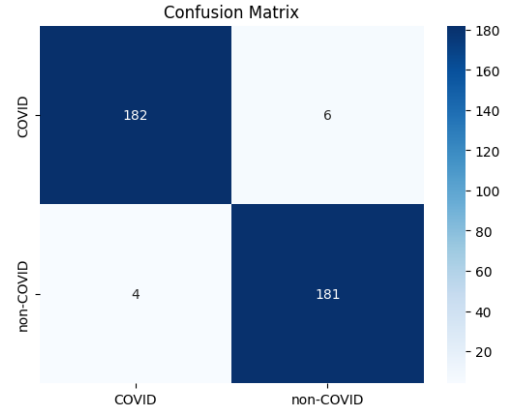


Fig. 3. Confusion Matrix evaluated on the 373 unseen test tensors.

Out of 373 total testing tensors, the system correctly identified 182 true positive COVID-19 cases and 181 true negative Non-COVID controls. Only 6 true COVID cases were misclassified as non-COVID (false negatives), and only 4 non-COVID scans were misclassified as positive infections (false positives). This low error profile highlights the model's high diagnostic sensitivity.

V. DISCUSSION

The quantitative performance achieved by this ResNet-50 framework proves the efficacy of deep residual topologies in computer-aided medical imaging. When compared to shallow architectural approaches such as LeNet-5, which often plateau around lower accuracies on highly complex datasets, our model's deep architectural design allows for optimal extraction of subtle clinical feature variations. Shallow models lack the

parameters and deep layer structures required to differentiate overlapping grey-scale visual distributions, frequently resulting in high false-negative rates when processing diffuse lung consolidations.

Our fine-tuning approach strikes a highly stable equilibrium between feature retention and cross-domain adaptation. Freezing stages 1 to 3 preserved generic low-level edge detection capabilities, while unfreezing the advanced layer4 residual block allowed the model to fine-tune its deeper filters to capture the specific multi-lobar textures characteristic of pulmonary infections. This targeted adaptation is confirmed by the highly symmetrical class-wise F1-scores (0.97 for both categories), indicating that the network is highly robust against minority class degradation without requiring complex dataset balancing techniques.

From a clinical triage perspective, the achieved binary AUC-ROC of 0.9929 holds significant diagnostic importance. It indicates an exceptionally low statistical probability of ranking a healthy scan above an actively infected slice. In practical deployment scenarios, the model's low false-negative rate (6 cases out of 188) is vital for minimizing missed infections, helping to limit inadvertent transmission within clinical wards. The rapid convergence observed within 12 epochs further emphasizes that using transfer learning with deeper networks achieves high diagnostic performance with minimal computational training overhead.

VI. CONCLUSION AND FUTURE WORK

This project successfully implemented a deep learning classification framework using a fine-tuned ResNet-50 architecture to detect COVID-19 from chest CT scans with an overall accuracy of 97.32% and an AUC-ROC of 0.9929. The experimental results prove that transfer learning combined with deep residual configurations provides a highly reliable approach for automated radiographic analysis, surpassing classic shallow network baselines.

Future work will focus on expanding the classification head to support multi-class classification task variants, allowing the network to differentiate COVID-19 from community-acquired bacterial pneumonias and chronic obstructive pulmonary diseases. Furthermore, we intend to integrate explainable AI layer modifications—specifically Gradient-weighted Class Activation Mapping (Grad-CAM)—to visualize attention distributions and highlight the exact pathological regions driving the classification decisions. Finally, we plan to convert the model to an optimized ONNX runtime to enable real-time inference within web applications for clinical triage settings.

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